Delayed but Growing Returns to a Selective Public University

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Setting: Public Flagship University in Colombia

Data

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Main Results

Mechanisms

Marginal Value of Public Funds

Robustness

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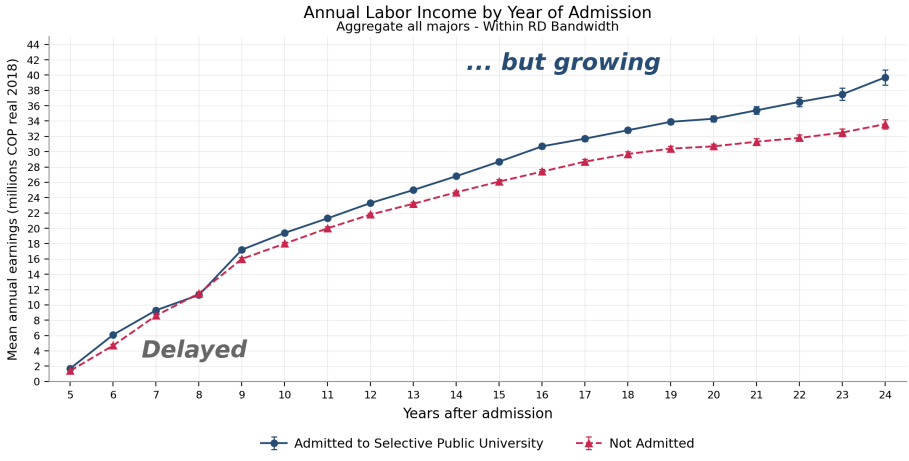
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Labor Income Trajectories in the Long Run (descriptive)



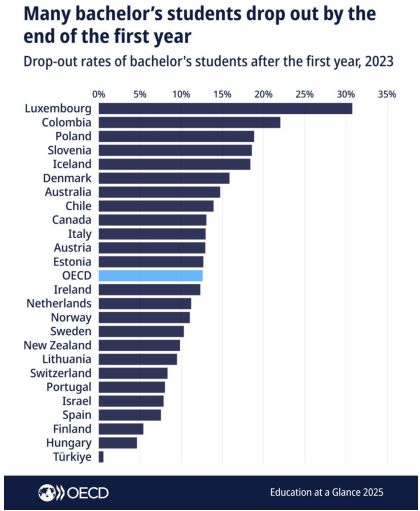
Note: Earnings reported in real 2018 Colombian pesos. 2018 PPP conversion factor: 1,322 COP per international dollar (World Bank, ICP).

Motivation 1: Credible Causal Evidence on College Returns Is Hard to Obtain, and Tends to Be Measured Too Early

- **Selection bias makes observational estimates unreliable**. Credible identification requires quasi-experimental designs — and these are difficult to construct at scale.
- **A growing RDD literature uses admission cutoffs to identify returns to college quality** (Hoekstra, 2009; Anelli, 2020; Jia & Li, 2021; Duryea et al., 2023, Zimmerman, 2014; Goodman et al., 2020).
- **The most comprehensive evidence to date — Mountjoy (QJE, 2026)**: 35 public universities in Texas. Finds +8% in earnings at 8–12 years. But this horizon reflects data availability, not necessarily when the return fully materializes.
- **Eight to twelve years is not enough for public universities in developing and middle-income countries**. Structural delays in graduation mean that study closing at year 10 is measuring students still finishing their degrees. Perhaps the return fully materialize several years later.

Motivation 2: The Delay Is the Rule, Not the Exception

First-year dropout rates exceed 20% at bachelor's level in several systems, including Brazil, Colombia, the French Community of Belgium, Peru and Romania.



Motivation 2: The Delay Is the Rule, Not the Exception

- **Graduating "on time" is rare — everywhere** (OECD, Education at a Glance, 2025; MEN–SPADIES, 2023). OECD: only **43%** of bachelor's entrants graduate within the theoretical duration; a third of those who eventually graduate need at least three additional years (43% on time vs 70% after +3 years).

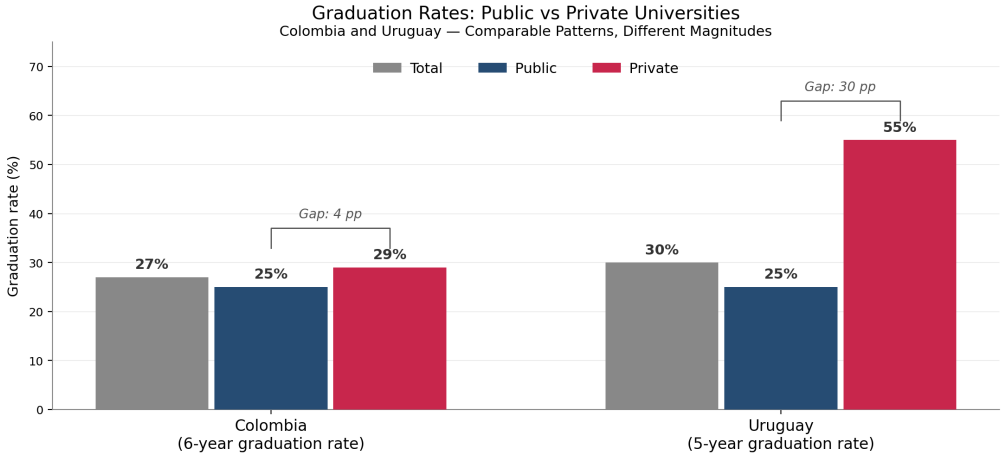
- **In Colombia, delays are more severe.** The 6-year graduation rate in public universities is 25% — well below private institutions (29%) and the U.S. benchmark (63%); Uruguay (30%).

- **The gap is structural, not accidental.** Public universities disproportionately serve low-income students who work while studying and have fewer financial buffers to stay continuously enrolled.

- **Implication:** when the evidence window closes at year 8–10, public university graduates in middle-income countries are still finishing their degrees — not yet revealing their returns.

Motivation 2: The Delay Is the Rule, Not the Exception

Colombia Vs Uruguay



Notes: Both indicators measure the share of first-year entrants who completed any tertiary degree by the indicated horizon. Comparison windows differ (Colombia: 6 years; Uruguay: 5 years) and metrics are not perfectly aligned. Colombia: MEN-SPADIES, cohort entering ~2017, public/private split for university tertiary education. Uruguay: Anuario Estadístico de Educación 2022 (MEC), as reported by El Observador (Jan 2024); cohort entering ~2017. UDELAR concentrates ~84% of enrollment in Uruguay; total is enrollment-weighted approximation.

Motivation 3: Early-Career Measurement Creates a Puzzle — That Longer Windows Resolve

- Riehl et al. (2016) rank Colombian universities on two dimensions simultaneously: learning value-added (exit exam scores) and earnings value-added (initial wages).
- The puzzle: Selective public universities lead on learning, while elite private universities lead on earnings. **How can a university produce more human capital yet appear weaker on returns?**
- The resolution is not institutional — it is about timing. An earnings ranking built on wages at or near graduation measures who enters the labor market fastest, not who accumulates the most human capital.
- It also affects the **Marginal Value of Public Funds**.
- When the evidence window is long enough, the problems dissaper. The return to the selective public university is not small — it is simply unmeasured when the window closes too early.

Motivation 4: Why Would a Return Exist? The Counterfactual Deteriorates Every Year

- **College human capital is productive** (Becker, 1964; Mincer, 1974). A selective public university generates a larger human capital stock than the available alternatives.
- **The counterfactual is not a fixed benchmark**. Rejected applicants sort into lower-quality universities (15%), short-cycle technical programs (10%), or exit higher education entirely (55%).
- **Each of these paths implies a lower and slower trajectory of human capital accumulation**. The longer the career, the larger the relative advantage of having attended the selective institution.
- The return does not just exist — it widens mechanically over time, as the gap between what the treated and control groups have accumulated continues to grow with every passing year in the labor market.

Motivation 5: Why Would the Return Keep Growing? Labor Market Dynamics Amplify the Advantage

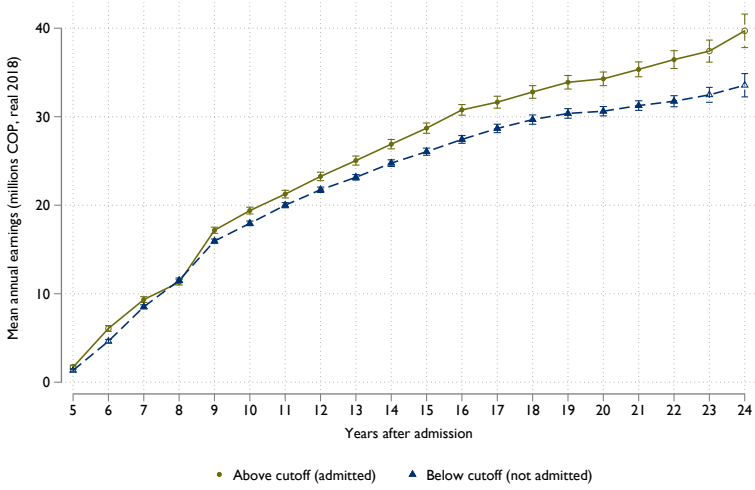
- **Skills beget skills** (Ben-Porath, 1967). A higher initial human capital stock raises the return to each additional year of on-the-job learning. Selective university graduates accumulate formal labor-market experience more quickly.
- **Better firms reward skills more steeply**. Selective university graduates are more likely to hold salaried positions in more productive employers. The extensive margin advantage is stable, but within those jobs, wages grow more sharply with tenure.
- **Employers learn over time** (Altonji & Pierret, 2001). At entry, firms observe the diploma but not true productivity. As employers learn, compensation converges upward — producing a return that is modest early and large late. This mechanism operates on the wage level, not on whether someone is employed.
- These mechanisms are consistent with a return that is near zero in early years, positive and growing through mid-career, and large in the long run.

This Paper

We estimate the return to a selective public university up to 24 years after admission — among the longest earnings windows in this literature.

- **Novel administrative dataset.** Universe of formal-sector earnings (PILA), bank lending records, and educational enrollment for applicant cohorts 2000–2004, tracked up to 2024.
- **Fuzzy regression discontinuity design.** We exploit program-specific admission cutoffs at Universidad del Valle — a highly selective public flagship in Colombia.
- **Long-run outcomes.** Annual earnings, cumulative labor market experience, and credit market access — consumption, automobile, and housing loans — as indicators of permanent income gains.

Preview of Findings: Earnings Trajectories in the Long Run



- No differences in early years for the marginal students around the cutoff. The gap in favor of the marginally admitted group widens over time.
- Further in the future, we show that these estimates can be significantly higher, underscoring the importance to measure returns with even longer periods.
- By years 16–18, admitted students earn approximately COP $4–5 million more per year, corresponding to a roughly 20–25% relative effect.

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Setting: Public Flagship University-Univalle

- **Colombia**
  - Population: 50 million (39 million in 2000). Students in tertiary education 2.5 million.
- **Cali, Valle del Cauca**
  - Population Valle del Cauca (state): 4.6 millions (3.4 millions in 2000). Students in tertiary education 184,000.
  - Population Cali 2.9 millions (capital 3th largest city in Colombia) (1.9 millions in 2000). Students in tertiary education 130,000.
- **Universidad del Valle**
  - The third largest University in Colombia, usually ranking 3 in Colombia according to international rankins (U- Sapiens or similar).
  - Undergraduate students 28,000, short cycle 5,000, posgraduate 2,000 (in 2000 Undergraduate students 18,000, short cycle 3,800, posgraduate 1,500).
  - Most important University on the south part of Colombia.

Setting: Public Flagship University-Univalle

- **Large Selective University.**
  - Most selective in the region: 32% admission rate.
  - Tuition subsidized: flexible fee scheme (charged based on family income).
- **Alternatives in the region.**
  - Certified universities: less selective but still quality certified.
  - Other higher education institutions: two-year programs, other non-certified universities.
  - No higher education.
- **Admission based solely on the national high school exit examination (ICFES).**
  - All Colombian high school graduates are required to take this examination.
  - Applicants apply to a single major per admission cycle; the cutoff is program-specific.
  - Admission decisions are publicly announced, with applicants informed of whether their score falls above or below the program cutoff.

Admission Process to Public Flagship University-Univalle

- **Admission process**
  - Each semester applicants can submit applications for the programs at the University
  - Each program assigns a different weight to the Saber11 score, according to the field. Example: higher weights to math for engineering.
  - The university calculates a new weighted score based on the Saber 11.
  - With the new score, each programs sorts the lists of applicants , admitting students up to the total seats available.
  - The number of seats available determines the cutoff for admission.

- **Score standarization**
  - We standardize the weighted score for all programs around 0, calculating the distance to the admission cutoff for each program.

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For applicants to the Selective Public University between 2000 and 2004, we construct a panel data structure tracking their academic trajectories, formal-sector earnings profiles, and credit histories up to 24 years after admission.

- **Admission Cohorts**: 2000-2004. Publicly available list of applicants by program and admission cycle.
- Records include the score assigned, score cutoff and admission results.
- Standardized test: End-of-high school test (Saber 11): near universal take-up for high school seniors. Standardized tests administered by the National Goverment.

Data Sources

- Enrollment and graduation: national registry of higher education system.
  - **National System for Dropout Monitoring in Postsecondary Education (SPADIES)**.
  - All postsecondary institutions in the country are mandated to report each semester to the Ministry of Education, registration, enrollment, graduation and dropout status for each student.
- **Administrative records on contributions to the social security system (PILA)**.
  - Monthly earnings reported by employers for formal wage earners, and reported by self-employed workers.
  - Records have national coverage for the universe of formal workers making health and pension contribution to the social security system.
  - Records from years 2008 to 2024, to reconstruct over 24 years of earnings trajectories.

Data Sources

- **Lending Operations reported to Financial Regulator (Format 341)**
  - Every bank or lending institution in the country is mandated to report all lending and loan operations to the Superintendencia Financiera de Colombia.
  - Each record register the borrower identity, the type of credit line, credit scores, loan size, interest rate among others.
  - Records from years 2005 to 2024, to reconstruct up to 24 years of credit history.
  - Specifically, we examine access to consumption credit, vehicle credit — as a means of financing durable goods purchases — and housing credit as a pathway to long-term asset acquisition.

Outcomes

- **Earnings and Labor Market**
  - **Annual Earnings**: Sum of all observed wages and earnings reported by year for an individual. When individual does not appear in the PILA, we record 0 formal earnings. Final variable in **real value 2018 Colombian pesos**, winsorized to 99 percentile.
  - **Formal worker**: binary variable recording whether the individual made at least one contribution in a year.
  - **Salaried worker**: binary variable recording whether the individual is working under a contract as employee receiving all the employment benefits beyond wages.

- **Credit Market**
  - Binary indicators for credit: binary indicators denoting whether an individual has any lending operation for consumption, vehicle, and mortgage in a given year.

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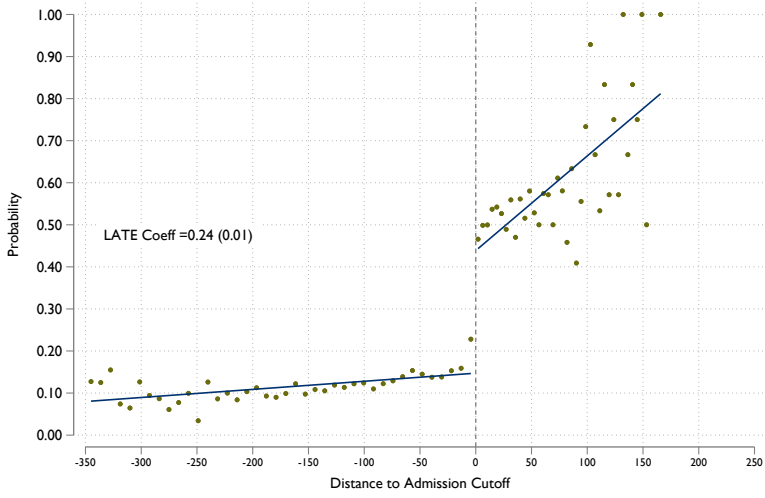
Robustness

Main Specification: Fuzzy Regression Discontinuity Design

$$Y_{ipc} = \beta_1 * A_{ipc} + \beta_2 * S_i + \beta_3 * A_{ipc} \times S_i + \theta_{pc} + \varepsilon_{ipc} \quad , i \in h$$

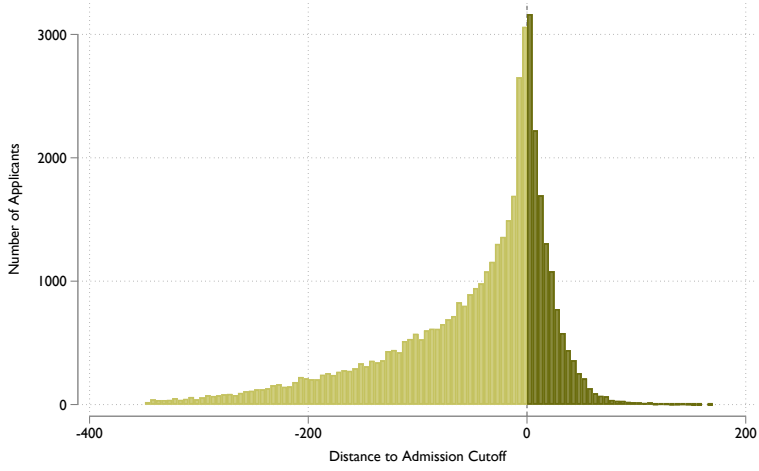
- Local Average Treatment Effects LATE β_1
  - A_{ipc} Attendance instrumented by admission binary indicator.
  - S_i Score assigned by the university, standardized score around 0.
  - Y_{ipc} Outcome variable for stacked c cohorts.
- RD Specification
  - Fixed Effects: p program, c admission cohort fixed effects.
  - MSE-optimal bandwidth h, asymmetric around cutoff using package rdrobust (Cattaneo et al.,2014).
  - Standard errors clustered at the individual level (Kolesar and Rothe, 2018).
  - Results are robust to other clustered standard errors (score level)

First Stage: Discontinuity on Enrollment



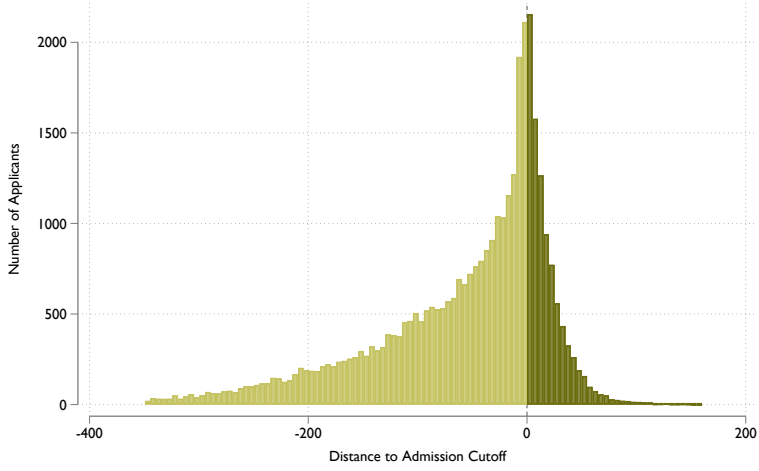
- In this setting, all students above admission cutoff are admitted.
- Clear discontinuity in enrollment of 24pp around the admission cutoff.

RD Validity: Continuity in Score Distribution



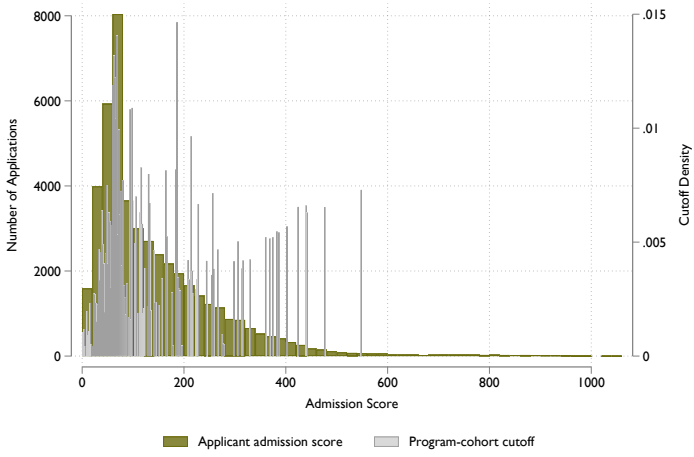
- Score distribution stacking all programs admissions. No statistically significant discontinuity of the score around the cutoff (RDDensity Test pval 0.874).

RD Validity: Continuity Holds on First Applications Only



- Applicants may apply in more than one admission cycle, so the pooled distribution contains repeat applications. Restricting to each applicant's **first** application removes any concern that re-application behaviour is itself a response to the cutoff.
- The density is equally smooth at the cutoff in this sample, so the main specification is not driven by repeat applicants.

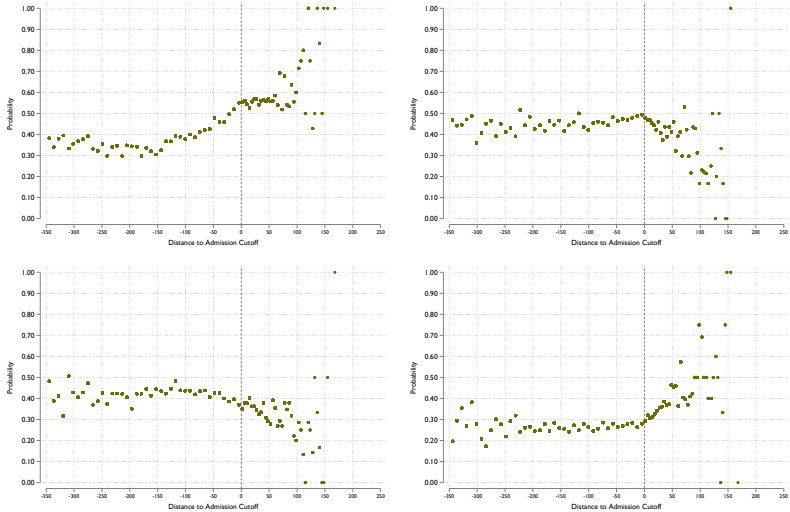
Where the Program Cutoffs Sit in the Raw Score Distribution



- The running variable is the *distance* to a program-specific cutoff. This figure shows the underlying raw weighted score together with the cutoffs themselves, weighted by the number of applications each cutoff governs.
- Cutoffs are spread across the interior of the score distribution rather than concentrated at one point: identification comes from many programs at many score levels, not from a single threshold.

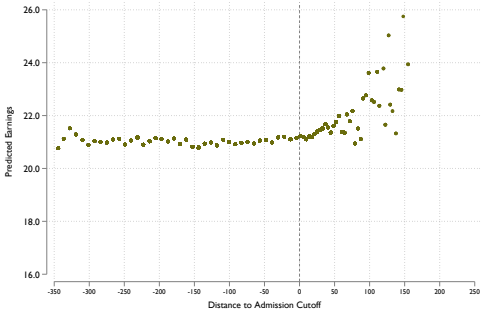
RD Validity: Covariate Balance

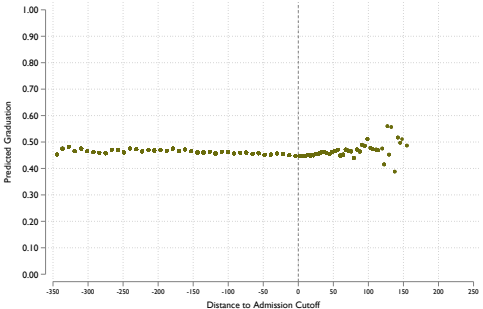
No discontinuity at the cutoff in covariates measured at admission



Gender, public high school, low income, and parents with a college degree. Age and the remaining parental-education and income categories are balanced as well

RD Validity: No Jump in *Predicted* Outcomes at the Cutoff





- Balance on covariates one at a time can hide joint imbalance. We therefore project earnings (left) and graduation (right) on the full set of admission covariates and test for a discontinuity in the **predicted** outcome.
- Neither index jumps at the cutoff: on everything observable at admission, marginally admitted and marginally rejected applicants were on the same expected trajectory. Whatever we measure later is not a pre-existing difference.

Sample characteristics

	All majors	
	All Applicants to Selective University	RD Sample Applicants within Bandwidth
	(1)	(2)
Observations	35,317	24,147
Demographics		
Male	0.49	0.52
Age at admission	17.5	17.6
Low Income	0.39	0.38
Public High School	0.47	0.48
Parents with college degree	0.29	0.29

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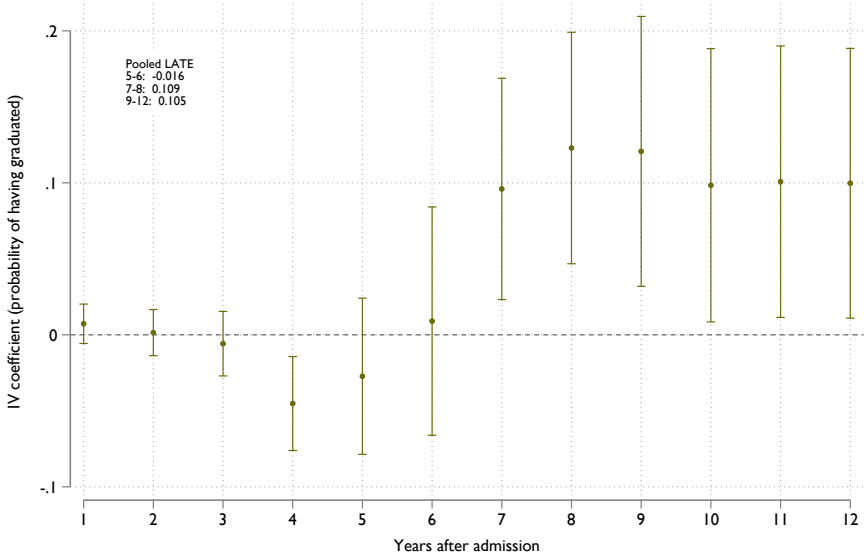
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Summary: Main Results labor market

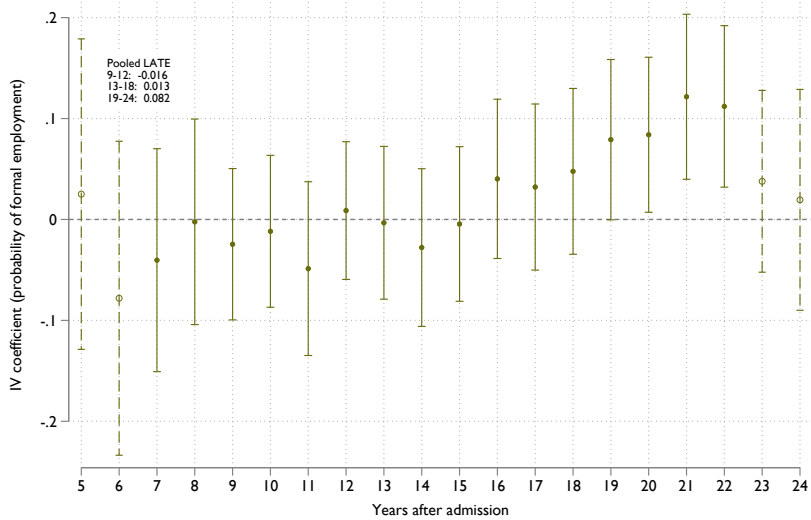
- **Enrollment and Graduation.** Large positive effect — stabilizes at +36 p.p. from year 9.
- **Formal Employment.** No effect on the extensive of the labor market.
- **Labor Income.** Delayed but growing. Near zero through year 8; +20% by years 9–12; +40% by years 16–18.

Graduation time tends to take longer for flagship Public University



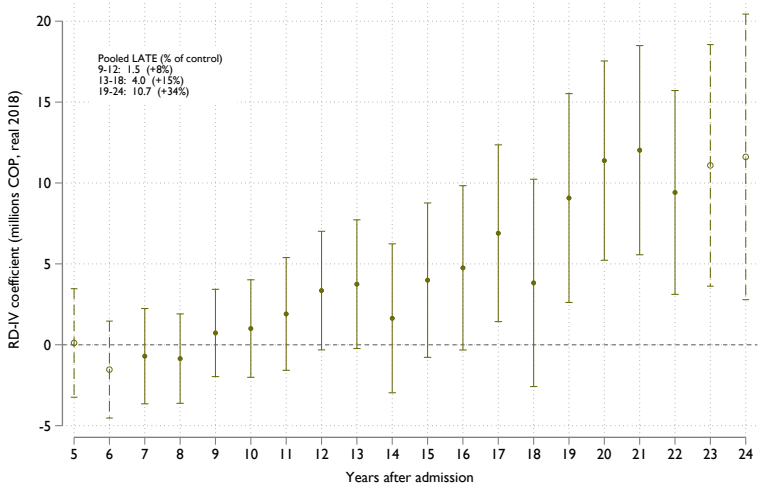
- No effect in graduation probability 5 to 6 years after admission

No long-term Effects on Formal Employment



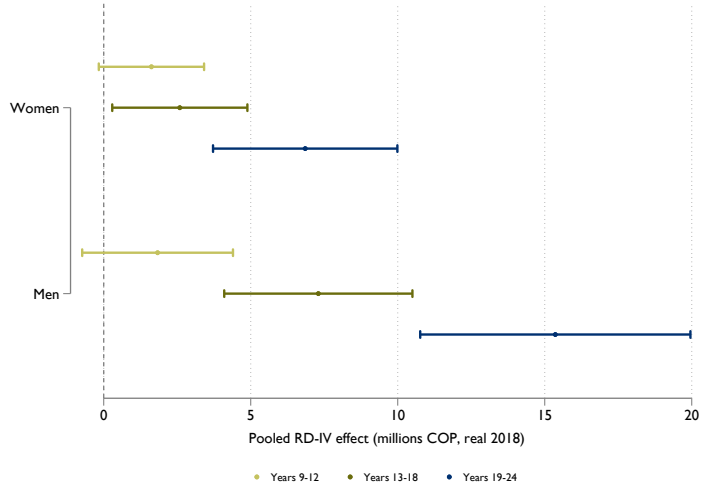
- 5 to 6 years after admission, marginally admitted students are likely to not be found having a formal employment, compared to the non-admitted.
- From 7 years after admission, there is no effect on the probability of having formal earnings.
- Control mean: 22% 5-6 years, 59% 13 years, 55% 22 years

Growing Returns in the Long Run



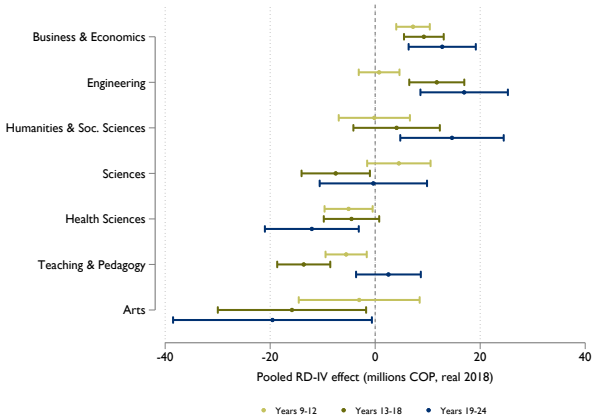
- Early years after admission (5-8): **No effect of attendance**
- **9 to 12 years** after admission: effect of attendance is **around 14% return**.
- **16-18 years** after admission: **around 28% return**.
- **23-24 years** after admission: additional premium **around 40% return**.

Growing Returns in the Long Run by Gender



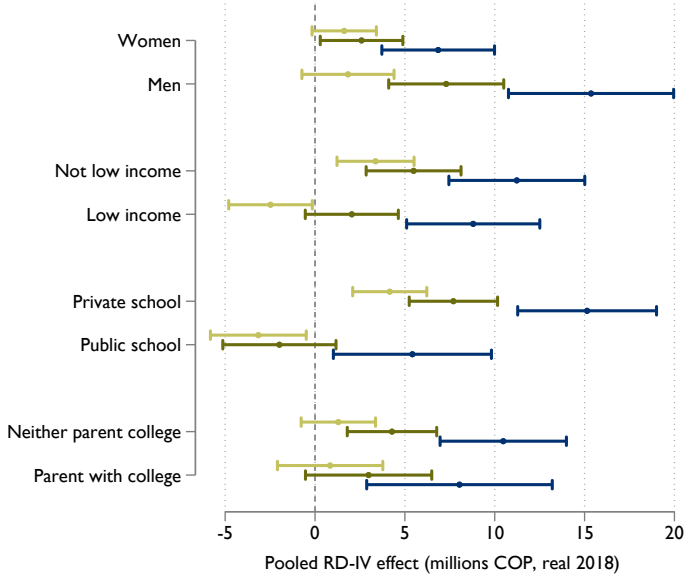
- Important differences by gender only emerge in the long-run

The Aggregate Return Is Concentrated in Business and Engineering



- Areas sorted by the pooled 9–24 effect; three period blocks per area, precision-weighted across years within each block.
- **Business & Economics** (+8.7M pooled) and **Engineering** (+6.2M) are positive and **rising**: Engineering runs 0.8 → 11.7 → 17.0M across the three blocks — the steepest gradient in the data.
- Health Sciences, Teaching & Pedagogy and Arts are negative throughout; Sciences is a flat zero.

Heterogeneity Across All Covariaes



Effects on Credit Markets

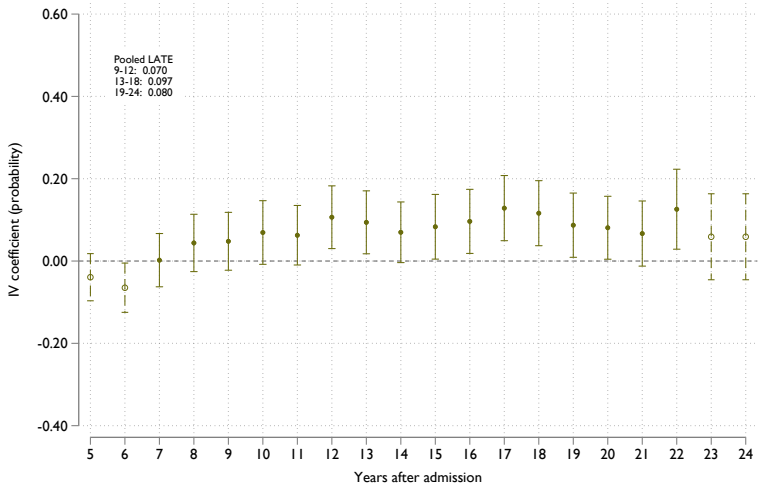
- **Credit Markets.** Positive effects on consumption and automobile loans, and on mortgage access late in the window — consistent with permanent income gains translating into real asset acquisition.

- **Consumption credit responds first and stays elevated**: around +10 pp from year 12 and +12 pp by year 22, tracking the earnings profile.

- **Automobile loans peak in the middle of the window** (+5.6 pp at year 18) and fade afterwards — a durable-goods purchase that happens once, not a permanent gap.

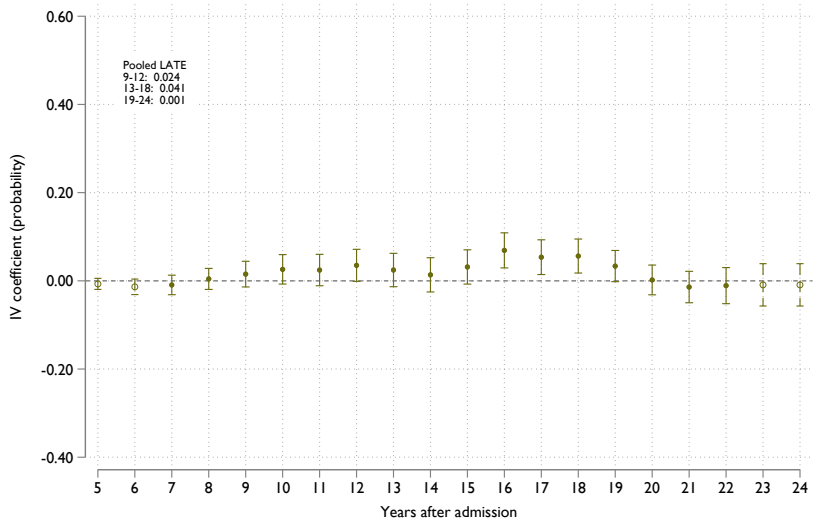
- **Mortgage access builds latest of all**: the effect is indistinguishable from zero before year 20 and reaches +7 pp by year 22 — consistent with housing as a long-term asset requiring sustained income gains before qualifying. This is visible only because the window runs to 24.

Postive Effects on Consumption Loans and Credit .



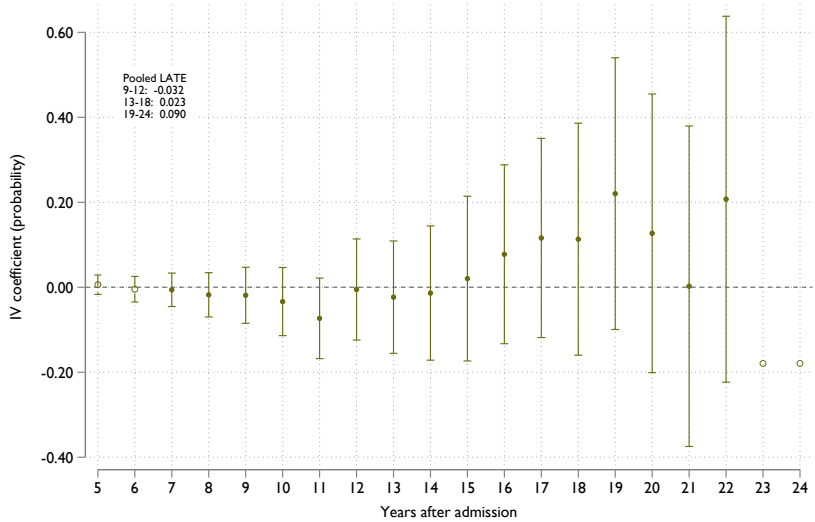
- Effects on having consumption loans (credit cards, auto loans) become positive and statistically significant from around year 12, following earnings effects starting to be positive.
- Point estimates are broadly flat thereafter — about +10 to +13 pp from year 12 through year 22 — in contrast to the steadily rising earnings effect. Access is an extensive margin that saturates; earnings are not.
- Control mean: 23% 5-6 years, 37% 13 years, 40% 22 years

Effects on Automobile Loans: Durable Consumption



- Positive effects on having an auto loan build through the middle of the window and peak at **+5.6 pp at year 18**, following earnings effects being positive.
- The effect **fades after year 20** (near zero at 20–24). An auto loan is a one-time durable purchase: the treated group reaches it earlier, and the control group partly catches up.

Effects on Mortgage Loans: Long Term Asset Acquisition



- Outcome is **cumulative years holding a mortgage** from admission to year z — a stock measure of how long the household has been able to sustain housing debt, not a one-year snapshot.
- The gap widens monotonically from year 15 onward, reaching about **0.2 additional years** by year 22, but it is **imprecise throughout** — zero is inside the

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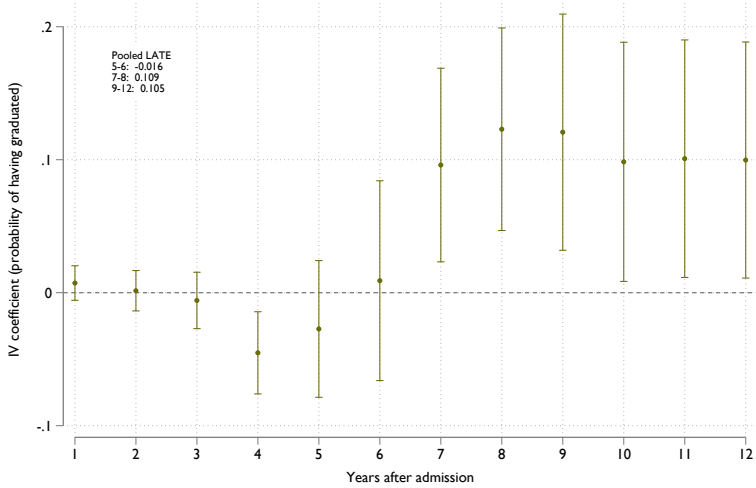
Robustness

Mechanism 1: The Delay Is Explained by Graduation Patterns

The marginal student at a selective public university takes longer to graduate — and therefore longer to enter the labor market with a degree.

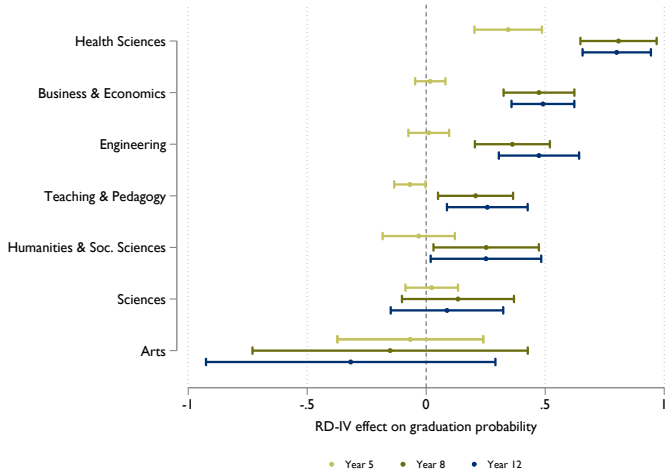
- On average, graduation rates at selective public institutions are lower than at private flagships — reflecting the higher share of low-income students who work while studying and have fewer financial buffers to stay continuously enrolled.
- The marginal admitted student faces additional barriers to timely completion, as they are the least academically prepared among their cohort.
- This explains why earnings effects are near zero in years 5–8: treated students are still enrolled — not yet in the labor market with a degree.

Graduation time tends to be longer for Public University



- No effect in graduation probability 5 to 6 years after admission.
- In years 7 to 8 the effect on graduation probability increases. From year 9 on the effect stabilizes around 36pp.
- Early career earnings effects reflect graduation barriers — not absence of returns.

Graduation Effects by Field



- Graduation effects at years 5, 8 and 12 by academic area: the **timing** of completion differs sharply across fields even where the year-12 effect is similar.
- Fields with the steepest earnings gradients also show the largest *late* graduation gains — the delay and the return share a common source.
- Caveat: this figure is built from the **legacy (pre-2024) graduation-by-area output** — it was never migrated to the current pipeline, so it is not directly comparable to the aggregate graduation figure above. Treat as indicative.

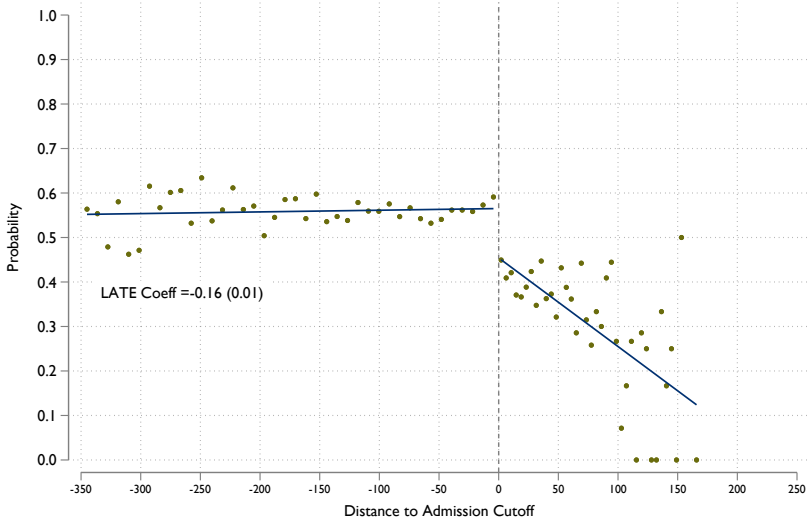
Mechanism 2: The Counterfactual Deteriorates Every Year

Where do rejected applicants go — and what does that imply for their earnings trajectory?

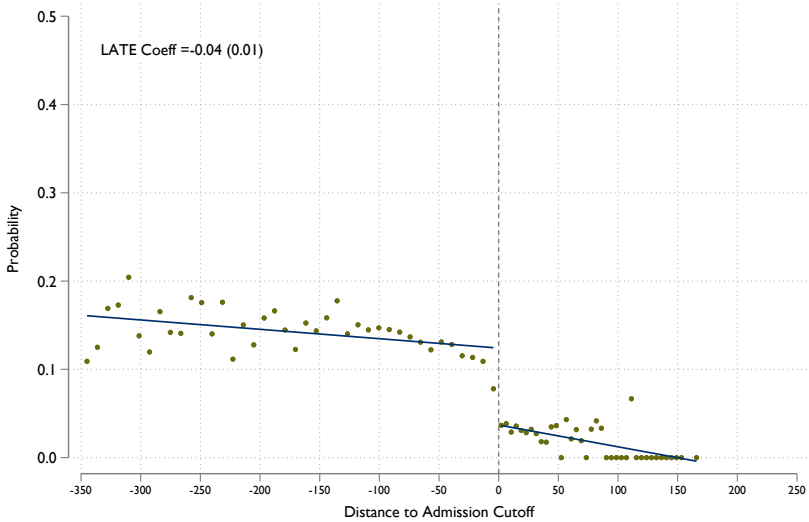
- **No higher education at all:** ∼55% of non-admitted students do not enroll in any postsecondary institution.
- **Lower-quality universities:** ∼15% enroll in non-accredited institutions.
- **Short-cycle programs:** ∼10% switch to two-year technical programs.

Each path implies a lower and slower trajectory of human capital accumulation — and a counterfactual that falls further behind every year.

Counterfactual Path 1: Most Non-Admitted Students Exit Higher Education

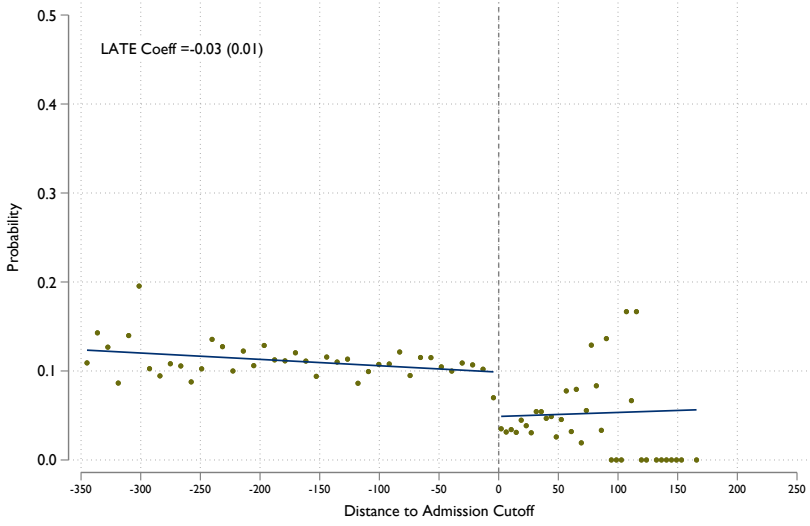


Counterfactual Path 2: Some Sort into Non-Accredited Universities

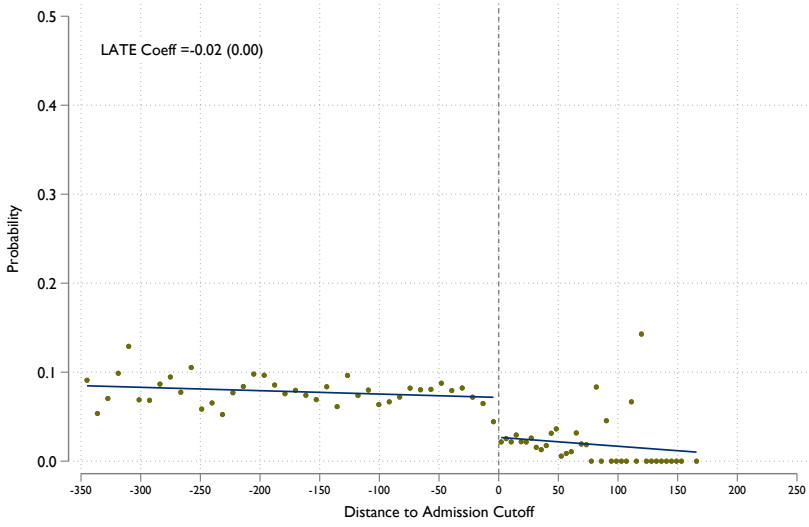


- About 15% in control group getting into college education with no certification of high quality universities.
- Estimated RD difference of -4pp on the probability of enrolling in low quality institutions.

Counterfactual Path 4: A Few Substitute to Another Accredited University

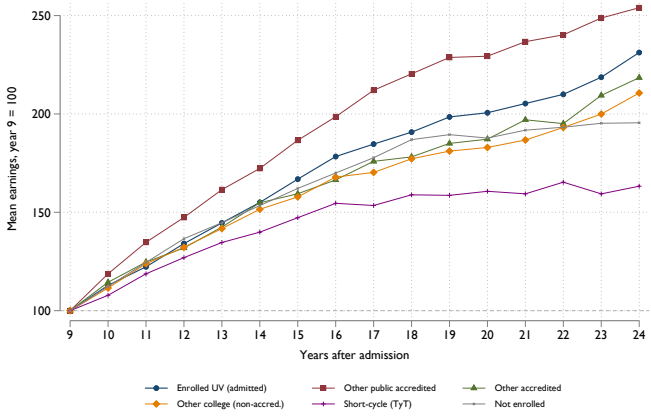


Counterfactual Path 3: Others Switch to Short-Cycle Programs



- About 10% in control group getting into two-year technical programs instead of five-years bachelor degree.
- Estimated RD difference of -2pp on the probability of enrolling in two-year programs.

Earnings Growth by Counterfactual Path (year 9 = 100)



- Mean earnings of marginal students by the alternative they actually ended up in, each series **indexed to its own year-9 level** so the figure compares *growth*, not levels.
- **The split is college vs. non-college, not selective vs. the rest.** By year 24 the four college groups stand at 211–254 on the index, while short-cycle (TyT) flattens at **163** and the not-enrolled group reaches **196**. TyT is essentially flat after year 16 (155 → 161 → 163).
- Univalle admits reach **231** — strong growth, but *not* the highest: that is the group that enrolled at another public accredited university (254).
- **Growth and levels rank differently.** "Other accredited" starts highest (21.2M vs. Univalle's 17.5M at year 9) and stays highest in levels (46.4M vs. 40.5M at year 24) while sitting mid-pack on growth.

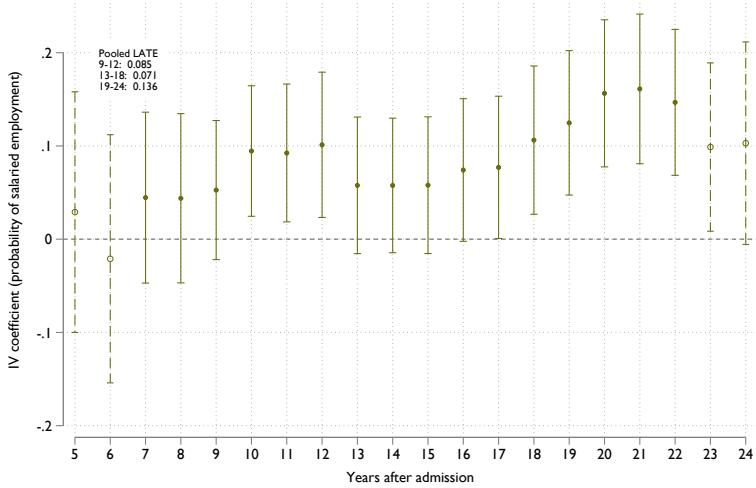
Mechanism 3: Labor Market Dynamics Amplify the Advantage

Beyond the counterfactual, two labor market channels can explain why the return keeps growing.

- **Firm quality sorting.** Do treated workers sort into salaried employment and larger, more productive firms — where returns to tenure are steeper?
- **Education-experience complementarity.** Is the return to each additional year of formal labor market experience larger for treated workers than for controls?

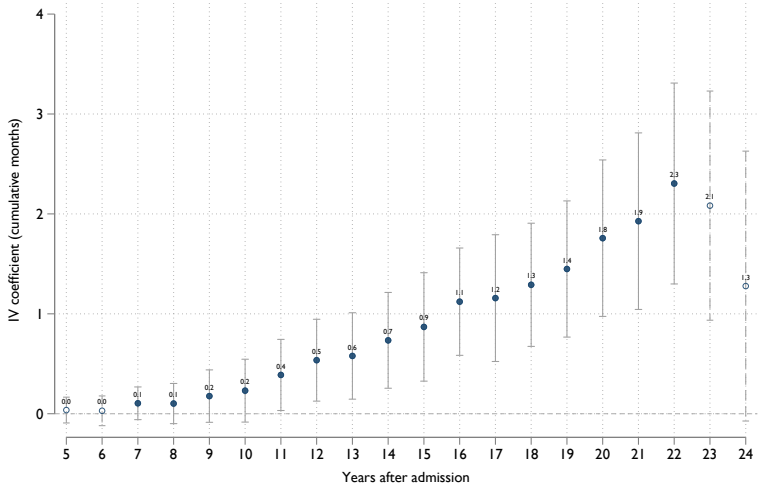
We now measure both channels **directly** from employer records in PILA: salaried status and cumulative experience, plus **firm size, firm tenure, and firm mobility** — data that is rarely available in this literature and is our main differentiation from Ng & Riehl (2024).

Attendants to Selective University more likely to be wage earners.



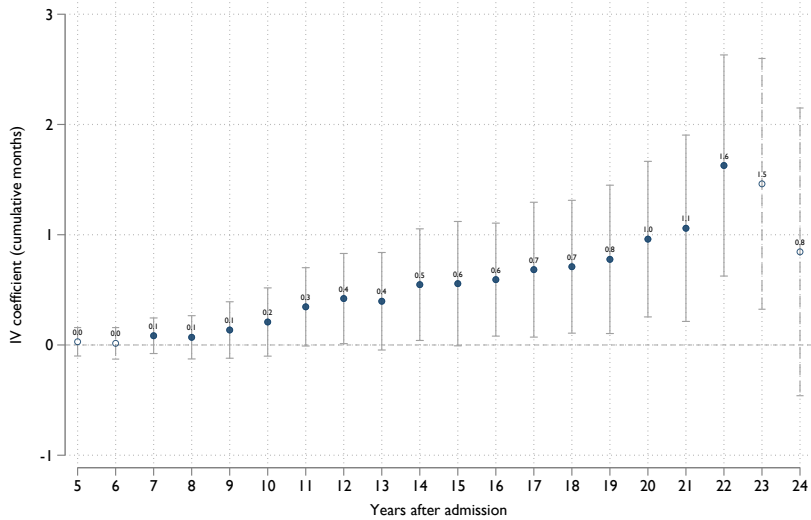
- In years 5–6, treated students are still enrolled — the control group (non-graduates) is already in the labor market as formal employees.
- From year 7 onward, the treatment effect on salaried employment is positive and **stable** at 5–12 p.p. (10–20% relative to control mean).
- A constant extensive margin advantage compounds into growing cumulative experience — the next slide.
- Control mean: 20% 5-6 years, 51% 13 years, 44% 22 years.

More cumulative experience in formal labor market



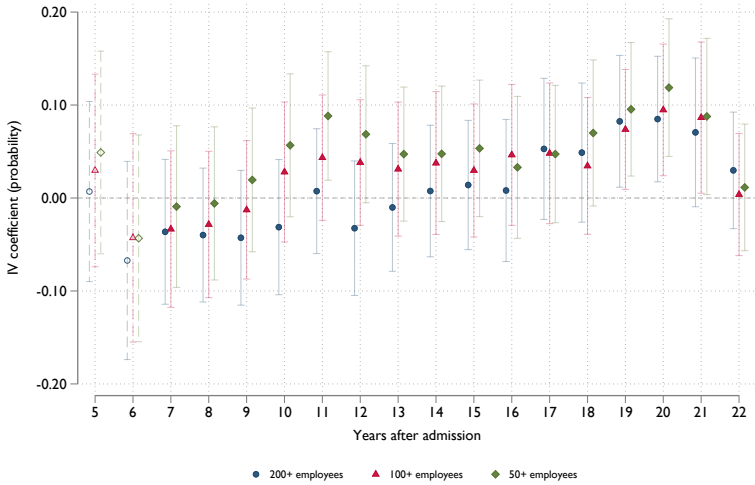
- Outcome is **cumulative years** with any formal employment since admission.
- The gap grows continuously — from near zero at year 8 to **+1.3 years by year 18** and **+2.3 years by year 22** ($t = 4.5$).
- Short vs. long-term differences are statistically significant, consistent with a compounding advantage over the career.

More cumulative experience as wage earners



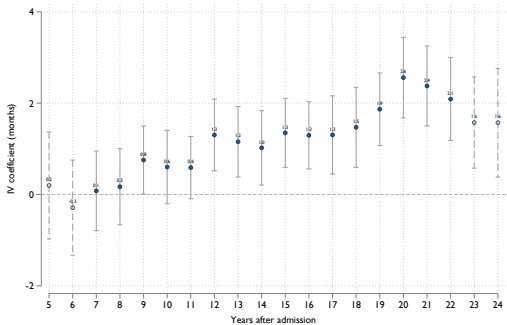
- Outcome is **cumulative years** as a salaried (wage-earning) worker since admission.
- A clear effect emerges from around year 12 (+0.42 years, $t = 2.0$), closely tracking the earnings pattern, and reaches **+1.6 years by year 22** ($t = 3.2$).
- Point estimates grow throughout the window: the extensive-margin advantage in *salaried* work compounds even though formal employment itself shows no

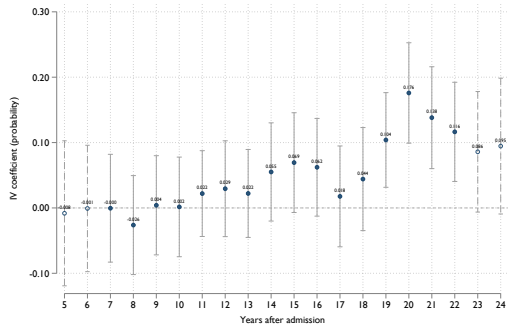
Sorting into Larger Firms — but Only After Year 15



- Probability of working at a firm above 200, 100 and 50 employees, on one axis.
- **The sign flips.** Effects are *negative* through year 12 (−3 to −7 pp at year 6–9: treated students are still studying while controls are already working), cross zero around years 14–15, and turn clearly positive late: at year 20, **+8.5 pp** (200+), **+9.5 pp** (100+) and **+11.9 pp** (50+), all significant.
- **Firm-size sorting is therefore a late-career phenomenon** — it appears exactly when the earnings gradient steepens, not when the first job is taken. This is evidence against a pure "better first job" story.

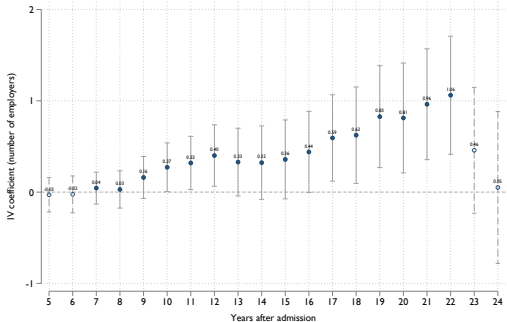
Longer Tenure with the Same Employer

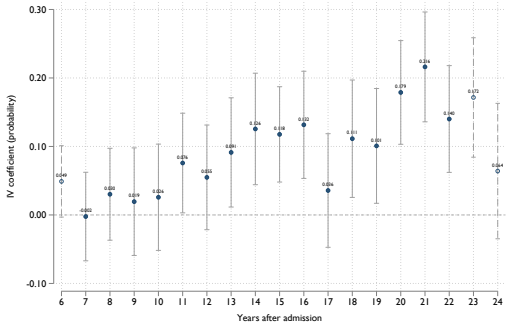




- **Left:** longest uninterrupted spell with one employer *within* the year (capped at 12 months). **Right:** probability that spell reaches at least 6 months.
- Both grow steadily and are precisely estimated: **+1.3 months at year 12**, **+1.5 at year 18**, **+2.6 at year 20** ($t = 5.7$). The probability of a $\geq$6-month spell rises **+17.6 pp by year 20**.
- Treated workers are not just employed more — they are **more continuously attached to a single employer**, which is the precondition for returns to tenure and employer learning.
- We report the *within-year* measure: the all-time tenure stock is left-censored at PILA's 2009 start by a cohort-varying amount, which mechanically inflates any "growing effect" reading. The within-year measure is immune to that.

More Employer Changes, Yet More Stable Attachment





- **Left:** cumulative number of distinct employers since admission — **+1.1 employers by year 22** ($t = 3.2$). **Right:** probability of being at the same firm as the previous year — **+17.9 pp at year 20** ($t = 4.6$).
- **These two go together rather than conflict.** Over a career the treated group accumulates *more* employer changes; in any given year it is *more* likely to have stayed put. That is the signature of **job-to-job moves up a ladder**, not of churn: each move is followed by a longer, more stable spell.
- Together with the firm-size and tenure results, this is the mechanism behind the growing earnings gap: treated workers reach larger firms, stay longer once there, and move deliberately between stable jobs.

Outline

Motivation

Setting: Public Flagship University in Colombia

Data

Methods and Validation

Main Results

Mechanisms

Marginal Value of Public Funds

Robustness

Marginal Value of Public Funds (Hendren y Sprung-Keyser, QJE 2020)

Policy evaluation asks a simple question: How much value do beneficiaries receive per peso of public spending?

The **Marginal Value of Public Funds (MVPF)** compares: (i) The **value of the policy** for beneficiaries. (ii) The **net fiscal cost** for the government

$$MVPF = \frac{\text{Willingness to Pay (WTP)}}{\text{Net Government Cost}}$$

Interpretation

- $MVPF > 1$: beneficiaries value the policy more than its fiscal cost
- $MVPF = 1$: value equals fiscal cost
- $MVPF < 1$: fiscal cost exceeds beneficiary value

Computing the MVPF: Flagship Public University

Step 1. Earnings effects of **RD-IV** up to 18 years after admission.

Step 2. Present value of lifetime earnings gains

$$PV(\Delta Y) = \sum_{t=1}^{H} \frac{\Delta y_t}{(1+r)^{t-1}}$$

Discount rate: $r = 0.03$, Tax rate: $\tau = 0.19$, Entry age: 17, retirement age: 60.

Step 3. Fiscal externalities from higher earnings

$$PV(\Delta \mathit{Taxes}) = \tau \cdot PV(\Delta Y)$$

Step 4. Net government cost: (i) Program cost. We compute the discounted operational cost per student-year over 5 years (PV(C)). (ii) $\mathit{NetCost} = PV(C) - PV(\Delta \mathit{Taxes})$

Step 5. Beneficiary willingness to pay: $\mathit{WTP} = (1-\tau) \cdot PV(\Delta Y)$

Step 6. Marginal Value of Public Funds: $\mathit{MVPF} = \frac{\mathit{WTP}}{\mathit{NetCost}}$

Marginal Value of Public Funds: using the 20-Year Returns

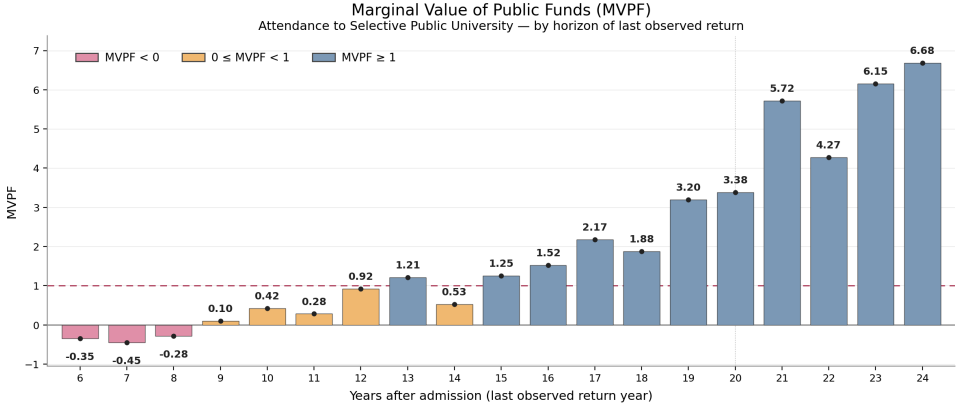
Component	Millions COP 2018
Present value of earnings gains, $PV(\Delta Y)$	110.78
Present value of additional taxes	21.05
Present value of program cost, $PV(C)$	47.58
Net government cost	26.53
Willingness to pay, *WTP*	89.73
MVPF	**3.38**

Interpretation: each additional peso of net fiscal cost generates about **3.4 pesos** of value for the marginal student induced to enroll.

These calculations use the 20-year return profile (RDD-IV) as a **conservative reference** (4 years before the last observed return). Discount rate 3%, tax rate 19%, retirement age 60, age at college entry 17, program duration 5 years. Earnings beyond year 20 are conservatively held constant at the year-20 estimate.

MVPF by horizon: from negative to large

The MVPF is highly sensitive to the assumed return profile.



Notes: Dashed red line: MVPF = 1 (breakeven). MVPF = WTP / Net Government Cost. Each bar uses the H-year return profile (RDD-IV), where H = 6, ..., 24. Earnings effects beyond H assumed equal to estimate at year H (conservative). Discount rate r = 3%, tax rate τ = 19%, entry age 17, retirement 60, program duration 5 years. Reference horizon (H = 20): MVPF = 3.38.

These calculations use the H-year return profile (RDD-IV), $H = 6, \ldots, 24$. Earnings effects beyond H assumed equal to estimate at year H. All other parameters as before.

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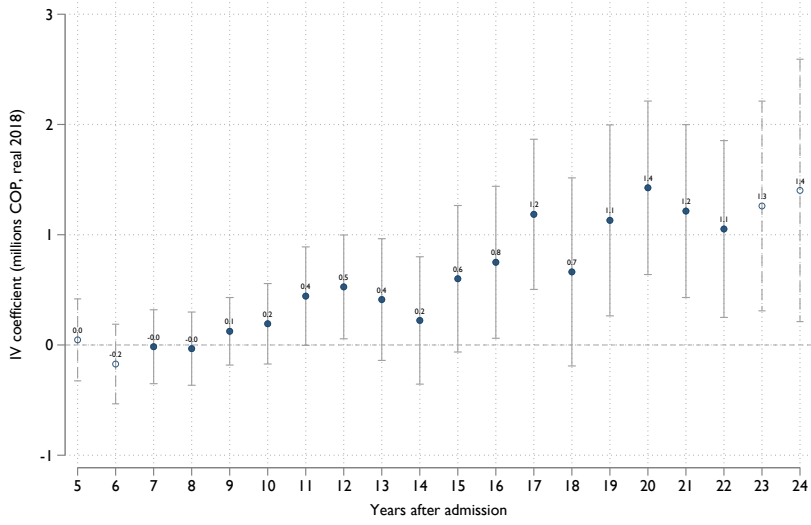
Main Results

Mechanisms

Marginal Value of Public Funds

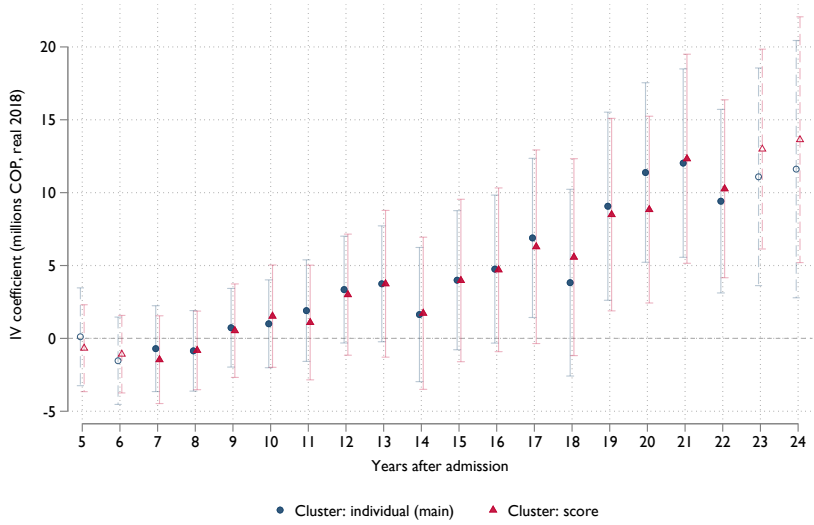
Robustness

Dynamic Effects of Attendance: Monthly Earnings



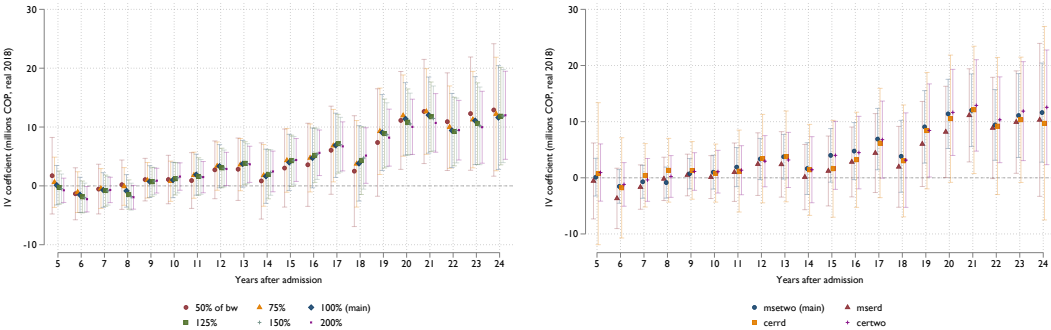
- The growing returns pattern is robust if we use other measures of income — here maximum monthly earnings rather than the annual sum, which removes any role for months worked.

Dynamic Effects of Attendance: Cluster



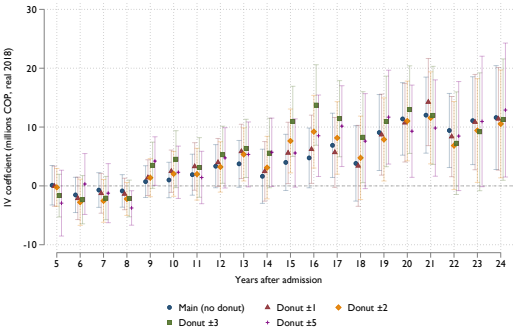
- Robustness to cluster by individual vs cluster by score

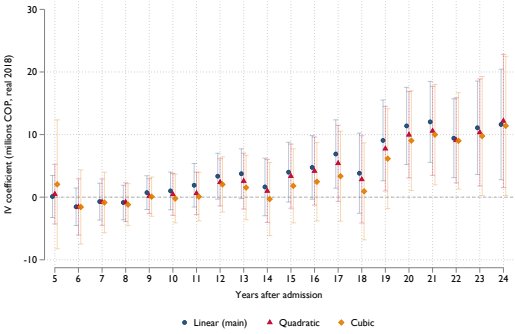
Robustness: Bandwidth Choice



- **Left:** the MSE-optimal bandwidth scaled from 50% to 200%. **Right:** alternative `rdrobust` selectors (`mserd`, `msetwo` — our main choice — `cerrd`, `certwo`).
- The delayed-then-growing profile is present under every bandwidth and every selector. Narrow bandwidths cost precision, as expected, but do not move the shape or the sign.

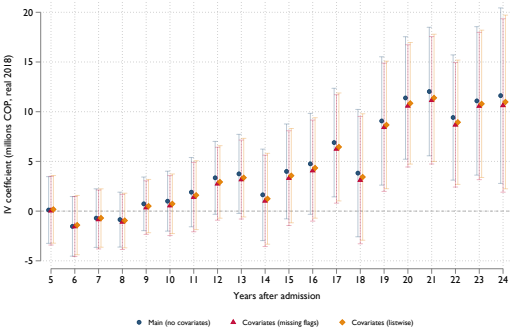
Robustness: Donut Hole and Polynomial Order

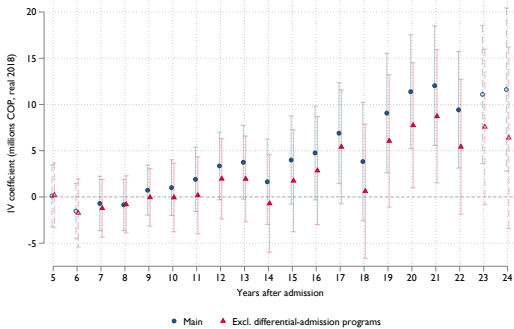




- **Left:** dropping applicants within $d = 1, 2, 3, 5$ score points of the cutoff. If scores were manipulated or cutoffs imprecisely recorded, the observations nearest the threshold would be the problem — removing them leaves the profile intact.
- **Right:** local linear, quadratic and cubic control functions. The result is not an artifact of functional form.

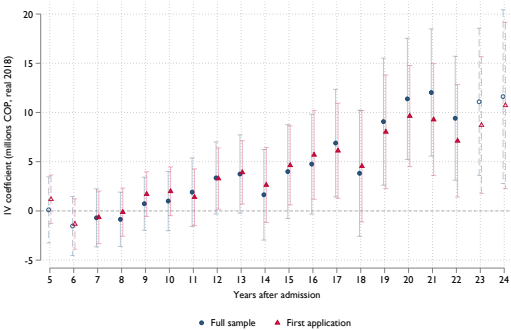
Robustness: Covariates and Program Exclusions

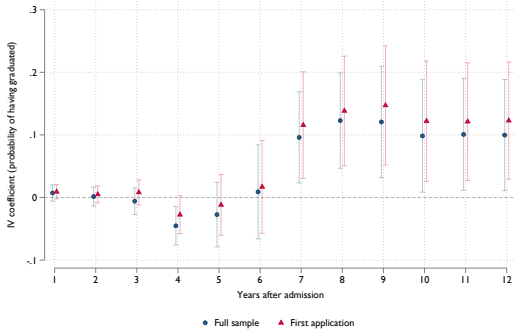




- **Left:** adding admission covariates to the main specification (with and without missing-indicator flags). Consistent with the balance tests, this changes little — which is what covariate balance predicts.
- **Right:** excluding program–cohort cells whose administrative cutoff differs from the one we reconstruct. The estimates are unchanged, so cutoff measurement is not driving the result.

Robustness: First Applications Only





- Main sample (all applications) overlaid with each applicant's **first** application only, for earnings (left) and graduation (right).
- Applicants rejected once may apply again, so the pooled sample contains repeat attempts and a rejected applicant may be admitted in a later cycle. Restricting to first applications removes that channel at the cost of sample size.
- Both the earnings gradient and the +36 pp graduation effect survive, with wider intervals. Credit and firm outcomes behave the same way (appendix).